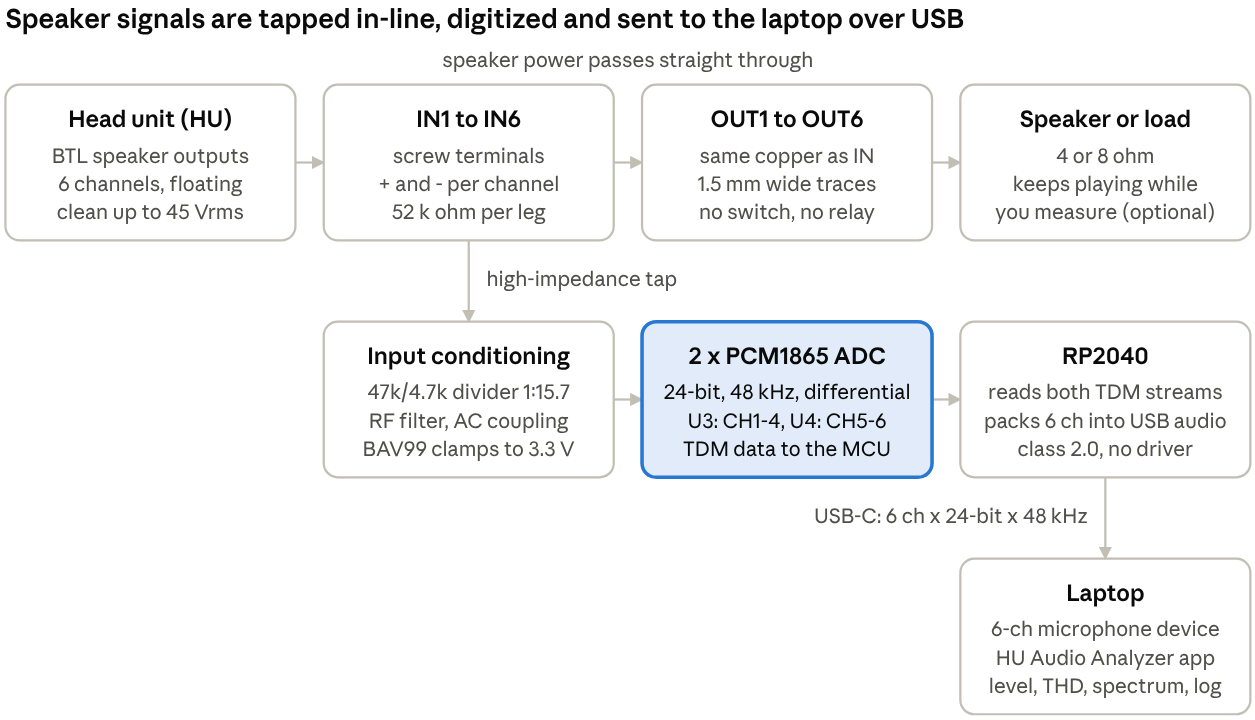
Digital Audio Analyzer User Guide

Oct 28, 2026

The Digital Audio Analyzer turns the six speaker outputs of a car infotainment head unit (HU) into a 6-channel USB audio input for a laptop, where the HU Audio Analyzer app measures level, frequency, waveform, spectrum, THD, THD+N and SNR per channel. This guide covers setup, wiring, measuring, calibration and troubleshooting for board rev A.

How the board works



signal chain · one of six identical channels

The top row is the speaker power path: each IN terminal is hard-wired to its OUT terminal, so the board only taps the voltage with a high-impedance divider and the speaker keeps working. The bottom path measures that voltage: the divider scales up to 66 Vrms down to the ADC's range, two PCM1865 chips digitize all six channels in sync at 24-bit/48 kHz, and the RP2040 streams them to the laptop as an ordinary USB audio input that the analyzer app reads.

Board tour

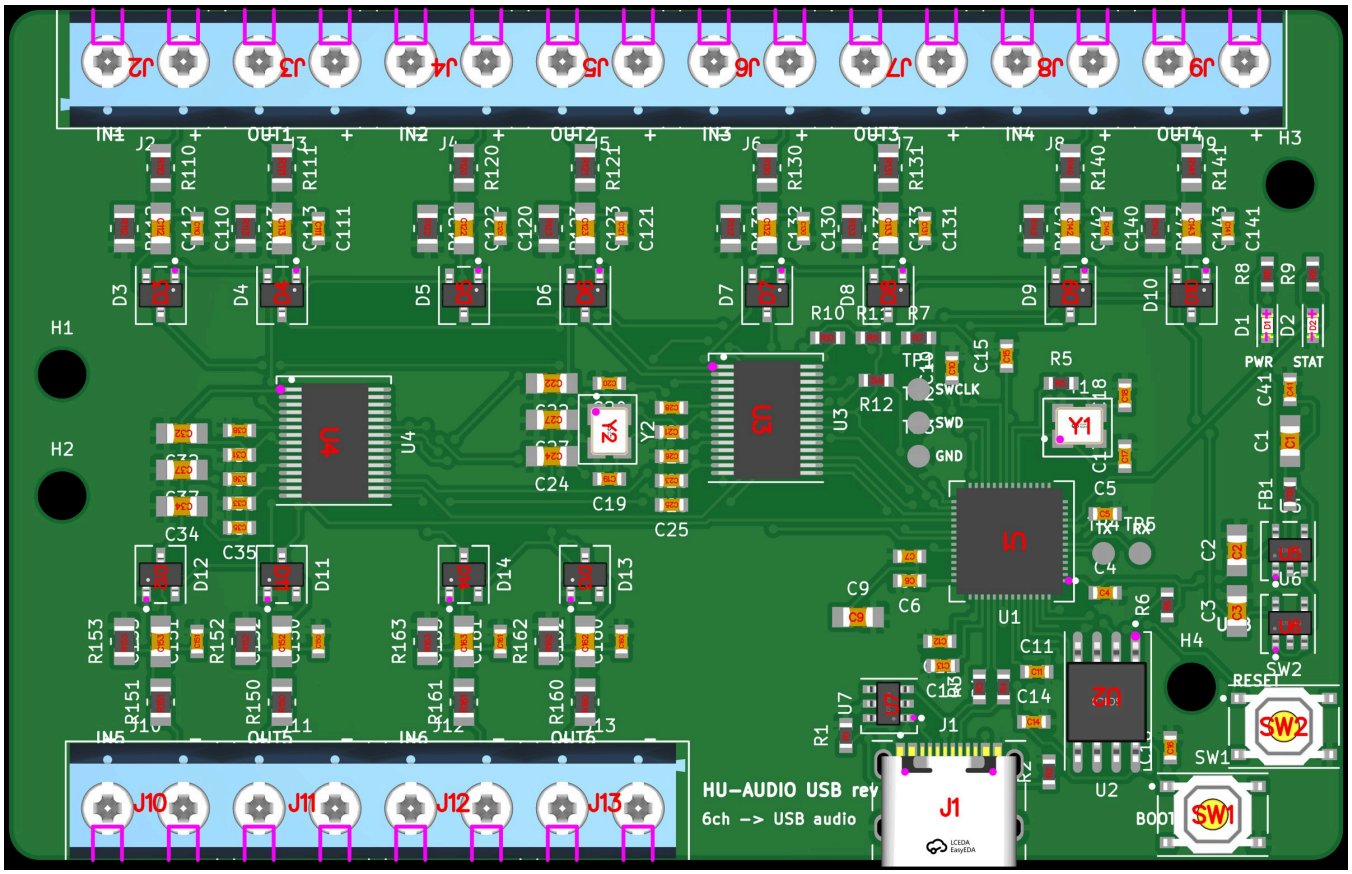
The board is 88 × 56 mm with the six input/output terminal pairs along the long edges and the USB-C connector at the bottom edge. Each terminal block is a 2-pole 5 mm screw terminal; the `+` and `–` marks are printed on the silkscreen next to each block.

Board photo: see page 3

Top side of the board: J2–J9 (IN1, OUT1 … IN4, OUT4) along the top edge, J10–J13 (IN5, OUT5, IN6, OUT6) at the bottom left, USB-C J1 at the bottom, BOOT (SW1) and RESET (SW2) at the bottom right, PWR and STAT LEDs at the right edge.

Item	Marking	Function
J2, J4, J6, J8	IN1, IN2, IN3, IN4 (top edge)	Channel 1–4 inputs from the HU speaker outputs
J3, J5, J7, J9	OUT1, OUT2, OUT3, OUT4 (top edge)	Pass-through of IN1–4 to the speakers (hard-wired, no switching)
J10, J12	IN5, IN6 (bottom edge)	Channel 5–6 inputs
J11, J13	OUT5, OUT6 (bottom edge)	Pass-through of IN5–6
J1	USB (bottom edge)	USB-C to the laptop: power (5 V, < 100 mA) and audio data
PWR LED	PWR	3.3 V present
STAT LED	STAT	Solid = streaming to the PC, slow blink = idle, fast blink (about 5/s) = ADC fault, about 2 blinks/s = safe mode after a firmware crash (unplug to clear)
SW1	BOOT	Hold while resetting to enter the firmware-update (RPI-RP2 drive) mode
SW2	RESET	Resets the RP2040
TP1, TP2, TP3	SWCLK, SWD, GND	Debugger pads (not needed for normal use)
TP4, TP5	TX, RX	3.3 V UART debug output, 115200 baud

Board photo



Top side of the board: J2–J9 (IN1, OUT1 … IN4, OUT4) along the top edge, J10–J13 (IN5, OUT5, IN6, OUT6) at the bottom left, USB-C J1 at the bottom, BOOT (SW1) and RESET (SW2) at the bottom right, PWR and STAT LEDs at the right edge.

Item	Marking	Function
H1–H4	MountingHole	3.2 mm holes for M3 standoffs (H4, next to RESET, is tight: use screw heads ⩽ 5.5 mm or nylon standoffs)

Main parts: U3 and U4 are PCM1865 4-channel 24-bit audio ADCs (U3 handles CH1–4, U4 handles CH5–6), U1 is the RP2040 microcontroller that packs the samples into USB audio, U2 is the 4 MB flash holding the firmware, U5/U6 are the 3.3 V regulators (digital and analog rails), U7 is the USB ESD protector.

Board capabilities

Measurement ranges for rev A as built (47 kΩ / 4.7 kΩ dividers, PGA 0 dB, 24-bit / 48 kHz). All six channels are sampled at the same instant, so phase and timing comparisons between channels are valid.

Quantity	Range / limit	Resolution / floor	Notes
Level (differential, IN+ to IN−)	0 to ≈ 66 Vrms (≈ 93 Vpk) full scale; clean up to ≈ 45 Vrms, above that the protection clamps start to soften peaks	≈ 0.4–0.6 mVrms noise floor (≈ −101 to −104 dBFS, measured); readings usable from ≈ 1.5 mVrms (HU volume step 1 reads about 0.5 dB high), within 0.1 dB above ≈ 5 mVrms	Vrms, dBV, dBu or dBFS; CLIP shown above full scale. A typical HU (1–12 Vrms) sits at −36 to −15 dBFS
Level accuracy	a few % uncalibrated	limited by your meter after Cal × calibration	The ADC's 10 kΩ input impedance is part of the divider, so calibrate once per channel
Lower ranges	PGA 0 to +40 dB via the `g <dB>` command: full scale down to ≈ 0.66 Vrms	—	Enter the new full-scale value in the app; useful for line-level sources

Quantity	Range / limit	Resolution / floor	Notes
Frequency response	≈ 2 Hz – 20 kHz flat (AC coupling corner 2.4 Hz, −0.13 dB at 20 kHz); analysis to 24 kHz	—	Both legs are matched, so common-mode hum cancels by ≈ 40 dB
Frequency readout	1 Hz – 24 kHz	≈ 0.1 Hz at 1 kHz (2.9 Hz FFT bins + parabolic interpolation)	Needs a steady tone
THD (harmonics 2–10)	0.003 % – 100 %	≈ 0.003 % (−90 dB) floor at HU levels	Covers the 0.05–1 % a head unit produces
THD+N	0.003 % – 100 %	same floor	Everything except the fundamental ±4 bins
SNR	up to ≈ 90 dB	—	Signal at −15 dBFS vs the ≈ −102 dBFS floor
Gain (relative to the reference channel)	±90 dB	≈ 0.1 dB repeatability	Both channels share the same divider design, so resistor tolerance cancels
DC offset	not measurable (DC removed)	—	Inputs are AC-coupled and the ADC high-pass filters, so this reads ≈ 0 V; check a HU DC offset with a multimeter
Input impedance	≈ 52 kΩ per leg (104 kΩ differential)	—	Negligible load on the HU
Speaker pass-through	≈ 6 A continuous	—	2 × 1.5 mm, 1 oz copper on both layers

Quantity	Range / limit	Resolution / floor	Notes
Absolute limits	≈ 45 Vrms / 65 Vpk differential	—	Never connect an input to chassis ground or +12 V while the laptop is chassis-grounded
Crosstalk (channel separation)	CH1 → CH2 ≈ −80 dB; all other pairs −85 to −127 dB	—	With 9 V on CH1, about 1 mV shows on CH2; the default Silent < level (1.2 mV) hides it

Safety and limits

The board is safe for any car head unit speaker output (full scale 66 Vrms; the protection clamps stay out of the signal up to about 45 Vrms, far above any head unit) and presents a light load of about 52 kΩ per leg, so the HU and the speakers behave exactly as if the board were not there.

- **No galvanic isolation.** The board ground is the laptop's USB ground. BTL speaker outputs are floating, so measuring them is fine, but never connect an IN terminal to chassis ground, battery +12 V, or a single-ended output that is referenced to chassis while the laptop is also chassis-grounded (charger plugged into the car). The simplest rule: run the laptop on battery while measuring in a car.
- **Speaker-level only.** Do not connect line-level RCA outputs expecting good resolution; they work, but a 1 Vrms signal is only −36 dBFS. Do not connect mains voltages or amplifier outputs above 40 Vrms.
- **Pass-through is passive copper.** OUTx carries the full HU power to the speaker; keep the screw terminals tight and the wire gauge the same as the car wiring (up to 1.5 mm² / 16 AWG fits the KF301 terminals).
- **Connect in this order:** HU off → wire IN/OUT → USB to laptop → HU on. Disconnect in the reverse order.
- The inputs are AC-coupled (4.7 µF) and clamped by BAV99 diodes to the 3.3 V rails; a DC fault on a speaker line is survived, but the clamps conduct, so switch the HU off and check the wiring if the app shows a large DC offset.

Laptop on a car charger

Using the board while the laptop is powered from the car works and damages nothing, but the measurements get noisier. Most car laptop and USB-C chargers are not isolated, so the laptop's USB ground, and with it the board ground, becomes the car chassis ground. Each HU speaker wire then feeds a small DC current into the board through the 47 kΩ + 4.7 kΩ divider (about 0.14 mA per wire at a 7 V BTL idle level); the 4.7 µF coupling capacitor keeps that DC away from the ADC, so levels, frequency and THD stay correct.

What changes: charger and alternator noise rides on the chassis ground that is now the measurement reference. The differential inputs cancel most of it (about 40 dB, limited by the 1 % divider resistors), so expect a 50/60 Hz or alternator line in the spectrum and somewhat worse THD+N and SNR readings. For clean numbers, do one of these:

1. Run the laptop on battery during the measurement.
2. Put a USB isolator (ADuM3160/ADuM4160 type, USB full speed) between the laptop and the board; this restores full galvanic isolation.
3. Use an isolated DC-DC laptop adapter.

Even chassis-grounded, the only way to stress the board is miswiring an IN terminal to battery +12 V, which pushes about 0.5 mA through the clamps and is survivable.

First-time setup

A new board needs the firmware once; after that it is plug-and-play on any Windows 10/11 PC (no driver, it is a standard USB Audio 2.0 device).

1. Flash the firmware

1. Plug the board into the laptop with a USB-C data cable. The PWR LED lights. With blank flash the RP2040 appears as a USB drive named **RPI-RP2**. If it does not: hold **BOOT**, tap **RESET**, release **BOOT**.
2. Copy `firmware\AudioAnalyzer_PC_fw.uf2` from the project folder onto the RPI-RP2 drive. The drive disappears, the board restarts, and the STAT LED starts a slow blink.
3. Windows now lists a recording device **Microphone (HU Audio USB Analyzer 6ch)**. Check Settings → System → Sound → Input.

2. Windows sound settings (once per laptop)

The app opens the board in WASAPI shared mode with Windows' **raw** option, which skips Windows' audio enhancements for the analyzer. With firmware v2.7 or later the Windows input-volume slider has no effect either: Windows sends it to the board, which ignores it. Check these once on every new laptop:

- **Audio enhancements off:** Settings → System → Sound → Microphone (HU Audio USB Analyzer 6ch) → Audio enhancements: Off. Also turn off Dolby, Waves or the laptop maker's audio tools for this microphone.
- **Microphone access on:** Settings → Privacy & security → Microphone: turn on Microphone access and Let desktop apps access your microphone. If either is off, Windows sends silence and every channel reads silent.
- **Format:** the board offers only 6 channel, 24 bit, 48000 Hz, so there is nothing to choose. If the app ever reports fewer than 6 channels, pick that format in the device's Advanced properties and click Reconnect.
- **Quick check:** feed a known signal into one input and compare the Level with your meter. If it matches, the laptop is ready.

3. Install the app

1. Install Python 3.10 or newer from python.org and tick **Add python.exe to PATH**.
2. Open the `pc_app` folder and double-click `run.bat`. The first run installs the packages (numpy, sounddevice, pyqtgraph, PySide6) and then opens the analyzer window.
3. Optional: run `build_exe.bat` once to get a single `dist\HU_Audio_Analyzer.exe` that needs no Python.

The window header shows the device it connected to, the host API, the mode (normally shared raw), the channel count and 48000 Hz. With nothing wired, all six channels show `silent` with a level of about 0.4–0.6 mV, the board's noise floor.

If the header ends with a NOTE, Windows did not accept raw mode; readings are still right as long as Audio enhancements stay off. Every connect writes `analyzer_connect_log.txt` next to the app: what Windows offered, what was tried and the board's status report. Send that file when asking for help.

To use another laptop, copy `dist\HU_Audio_Analyzer.exe` and `analyzer_settings.json` (Cal ×, channel names, Full scale, Silent level) to one folder on it. Nothing else needs installing; if Windows warns about an unknown publisher, choose

More info → Run anyway. Work laptops may block unknown USB devices or unsigned programs by policy.

Making connections

Each channel is an in-line tap: the HU speaker wire pair goes into **INx**, the speaker (or a dummy load) goes onto **OUTx**, and the board measures the voltage across the pair. INx+ and OUTx+ are the same copper, INx− and OUTx− are the same copper, so the speaker keeps playing while you measure.

Board terminal	Connect to	Note
IN1 + / −	HU channel 1 speaker + / − (e.g. front left)	top edge, leftmost block
OUT1 + / −	Speaker or load for channel 1	optional; leave empty to measure unloaded
IN2 … IN4	HU channels 2–4	top edge
OUT2 … OUT4	Speakers 2–4	top edge
IN5, IN6	HU channels 5–6 (e.g. rear or subwoofer)	bottom edge
OUT5, OUT6	Speakers 5–6	bottom edge
USB	Laptop USB-A or USB-C port	use a data cable, not a charge-only cable

Steps:

1. Switch the HU off (ignition off or HU unplugged).
2. Strip 6–7 mm of insulation, insert each wire fully into the terminal and tighten the screw. Tug-test every wire.
3. Keep polarity consistent: HU `+` wire to IN `+`, HU `−` wire to IN `−`. Levels and THD are correct either way; polarity only matters for the waveform display and for comparing phase between channels.

4. If you want a load without speakers, connect a 4 Ω or 8 Ω power resistor (rated for the HU power, e.g. 50 W) across OUTx. Measuring with no load is also valid for most HU outputs but shows the open-circuit voltage.
5. Plug in USB, start the app, then switch the HU on.

Which HU wire is channel 1 does not matter to the board; use the editable channel-name column in the app (click a name such as `CH1` and type `FL`) to label the channels the way the car is wired. Names are remembered between runs.

Taking measurements

Play a known test signal on the HU (a 1 kHz sine track at a fixed volume is the standard; a USB stick or Bluetooth test-tone file works) and read the table. Every value updates 10 times per second from the latest 0.34 s of audio.

Column	Meaning	Typical use
Level	True RMS of the AC signal, in the unit chosen at the top (Vrms, dBV, dBu or dBFS)	Output power: $P = V^2 / R$ (e.g. 8.9 Vrms into 4 Ω = 20 W)
Gain dB	Level of this channel relative to the channel picked in the Gain ref dropdown: $20 \cdot \log10(V / V_ref)$. The reference row shows ref; – when no reference is picked or a channel is silent	HU gain (test signal into a spare channel as reference), channel balance, fader and crossover attenuation
Peak	Highest instantaneous voltage in the block	Clipping onset: peak stops rising while volume goes up
Freq (Hz)	Fundamental frequency, interpolated to about 0.1 Hz	Confirms the test tone, finds hum (50/60 Hz)
THD %	Harmonic distortion, harmonics 2–10 relative to the fundamental	Amplifier linearity; rises sharply at clipping
THD+N %	Everything except the fundamental (harmonics + noise + hum)	The usual spec-sheet number
SNR dB	Fundamental power over the remaining noise power	Noise floor check with a clean tone

Column	Meaning	Typical use
DC	Average of the signal after the board's AC coupling and the ADC's high-pass filter, so it reads about 0 V even if the HU has a real DC offset	Not a DC-offset test: check HU DC offset with a multimeter on DC volts across the speaker pair, HU muted
Cal ×	Per-channel correction factor (editable)	See Calibration
Status	`ok`, `silent` (below the Silent < level in the top bar, default 1.2 mV; Freq, THD, THD+N and SNR then show –) or `CLIP` (ADC full scale reached)	Lower the HU volume if CLIP shows

Typical procedures:

1. **Maximum undistorted power.** Play 1 kHz, raise the HU volume step by step and log the Level and THD columns. The volume step where THD jumps from < 1 % to several percent is the clipping point; the Level just before it is the rated clean output.
2. **Channel balance.** Play the same tone on all channels (mono source, balance/fader centred) and compare Level in dB; differences above 0.5 dB point to an HU or wiring problem.
3. **Frequency response.** Play a sweep or stepped tones (20 Hz, 50, 100, 200, 500, 1 k, 2 k, 5 k, 10 k, 20 kHz) and read Level at each step; the spectrum plot shows the shape live.
4. **Noise and hum.** Pause the source at a fixed volume: Level shows the idle noise, the spectrum shows hum lines at 50/60 Hz and alternator whine as a moving peak.
5. **Crossover / fader behaviour.** Watch the six levels while moving HU fader, balance and subwoofer settings.

Measuring gain

The board only sees the HU outputs, so gain is derived from two level readings. Pick the reference in the **Gain ref** dropdown at the top (`none` turns the column off) and the Gain column of every channel updates live, also into the CSV log.

- **HU gain from an analog input (AUX, line-in, external amp input).** Feed a 1 kHz test signal into the HU and wire the same signal to a spare board channel, e.g. IN6. Select CH6 as Gain ref; CH1–CH5 now show the HU gain in dB directly. A 1 Vrms line signal

reads at about −36 dBFS, which is still 55 dB above the noise floor, so the result is accurate to about 0.1 dB (both channels share the same divider design, so resistor tolerance cancels).

- **Sensitivity from a digital source (USB stick, Bluetooth).** Play a file with a known level, e.g. a 1 kHz sine at −20 dBFS, at a fixed volume step. The Level column is the output for that input; quote it as "x Vrms at −20 dBFS, volume n" or convert to dB against your own 0 dB reference.
- **Balance, fader, crossover.** Select CH1 as the reference and play the same tone on all channels; the other rows show their offset in dB (balance should be within ±0.5 dB; a subwoofer channel with a low-pass shows its attenuation at the test frequency).

Gain uses the true-RMS level of each channel, so both channels must carry the same test tone; with music playing the column still updates but only means something if the channels carry the same program.

Use **Start CSV log** before a run to save one row per 100 ms with level, peak, frequency, THD, THD+N and SNR for all six channels; open it in Excel afterwards. **Record WAV** stores the raw 6-channel 24-bit audio for later analysis in Audacity, REW or Python.

The plots: the waveform shows the last 20 ms in volts (so a 1 kHz tone shows 20 cycles); the spectrum is in dBFS with a logarithmic frequency axis. The checkboxes under the plots hide channels you are not looking at.

Calibration

Uncalibrated accuracy is a few percent (1 % divider resistors, but the PCM1865's 10 kΩ input impedance, which is part of the divider, is only loosely specified). One calibration against a true-RMS meter brings it to the meter's accuracy.

How the scale works: the app converts digital full scale (0 dBFS) to volts with the **Full scale** value in the top bar, 66.0 Vrms by default for rev A (47 kΩ / 4.7 kΩ divider into the ADC's 10 kΩ input, PGA at 0 dB). Each channel then multiplies by its own **Cal ×** factor (default 1.0000). Both are saved in `analyzer_settings.json` next to the app.

Procedure (needs a true-RMS multimeter or an oscilloscope):

1. Play a 1 kHz sine from the HU at a medium volume into channel 1 (a steady 2–10 Vrms is ideal; 1 kHz is inside the AC range of every true-RMS DMM).
2. Measure the voltage directly across the IN1 screw terminals with the meter.
3. Read the app's CH1 Level in Vrms.

4. Set CH1 **Cal ×** = meter voltage ÷ app voltage (for example 5.12 / 5.05 = 1.0139). Press Enter; the Level now matches the meter.
5. Repeat for CH2–CH6 by moving the meter and the signal to each channel.
6. Optional sanity check: with the HU muted, Level should read below 5 mV; with the inputs shorted it should read below 1 mV.

If all six channels are off by the same ratio, change **Full scale** instead of the six factors. If the PGA gain in the firmware is ever changed, Full scale must change with it (each 6 dB of PGA halves the full-scale voltage).

Frequency, THD and THD+N need no calibration: they are ratios, independent of the scale factor.

Debugging

Start with the two LEDs, then the app header, then the board's own status report: click **Board status** in the app (no soldering, no extra cable: the board exposes a USB serial port next to the audio device, firmware v1.1 or later).

Symptom	Likely cause	What to do
PWR LED off	No USB power, charge-only cable, bad USB-C connector	Try another cable/port; check 5 V on C41 and 3.3 V on C1
PWR on, no USB device, no RPI-RP2 drive	RP2040 not running: crystal Y1, flash U2, or USB data lines	Hold BOOT + tap RESET; if still no drive, check 1.1 V on C4/C5 and 12 MHz on Y1
RPI-RP2 drive appears every time	Firmware not written, or BOOT button stuck	Copy the .uf2 again; check SW1 is not shorted
STAT LED fast blink	An ADC did not start: I²C problem, U3 crystal (24.576 MHz), or U4 PLL not locked	Read the UART log (below); check 3.3 V on U3/U4 pin 13/14 and the analog rail on pin 8
STAT slow blink but Windows shows no device	USB descriptor not enumerated	Check the Device Manager for a yellow mark; re-flash; try a USB 2.0 port or a hub

Symptom	Likely cause	What to do
App: Device NOT FOUND	Device name mismatch or not enumerated	Click Reconnect; verify the recording device exists in Windows Sound settings
App shows 2 channels only	Windows shared mode default format is stereo	Set 6 channel / 24 bit / 48000 Hz in the device's Advanced properties, click Reconnect
All channels `silent` with signal applied	Wrong terminal row, HU muted, wires not clamped	Check that the HU is playing (speaker sound), wiggle-test the screw terminals
One channel dead, others fine	Open BAV99 / divider resistor or a broken terminal on that channel	Measure the channel's 47 kΩ resistors (R1x0/R1x1) and the 4.7 kΩ (R1x2/R1x3)
Level reads half or double expected	Cal × or Full scale changed, or PGA edited in firmware	Reset Cal × to 1.0 and Full scale to 66.0 and recalibrate
THD high at low volume on every channel	Clock problem (U4 PLL unlocked gives noise bursts) or a loose ground	UART log: ADC2 must show `locked`; check hum at 50/60 Hz in the spectrum
CH5/CH6 distorted or swapped with each other	U4 not frame-aligned after a glitch	Click Reconnect (restarts the capture frame-aligned); power-cycle the board
Periodic clicks in the recording	USB packet loss (busy laptop, long hub chain)	Plug directly into the laptop; close other audio apps; use firmware v2.6 or later (older firmware dropped audio every 5 s and while Board status ran)
Large DC offset on a channel	Clamp diode conducting: HU DC fault or wire to chassis	HU off, check the speaker wires for DC with a meter

Symptom	Likely cause	What to do
Laptop sound stops (speaker icon blank or ×) when the board is plugged in	Firmware older than v1.9 hangs its USB when Windows starts the stream, and Windows audio hangs with it	Unplug, flash the current firmware, run pc_app\fix_windows_audio.bat (or reboot)
Readings follow the Windows input-volume slider (e.g. CLIP at 100)	Firmware older than v2.7: Windows applies its own software gain	Flash v2.7 or later; if the slider still acts, uninstall the microphone in Device Manager and replug
Every channel silent on a new laptop, though Board status shows streaming=1	Windows microphone privacy blocks desktop apps	Settings → Privacy & security → Microphone: allow access and desktop apps
STAT LED blinks about 2 times/s	Safe mode: the firmware crashed once and stopped streaming to protect the laptop	Click Board status and save the text (it holds the crash record), then unplug and replug to clear
CH2 shows a level or 1 kHz while only CH1 is driven	Crosstalk, about −80 dB from CH1 (≈ 1 mV with 9 V on CH1)	Normal; raise Silent < in the top bar if it shows as ok
App stays on "Device: connecting…" or shows an open error	Another analyzer window or program holds the COM port or the audio device	Close other analyzer windows, click Reconnect, read analyzer_connect_log.txt next to the app

Board status over USB (no soldering)

Besides the 6-channel microphone, the board enumerates a virtual COM port named **HU Analyzer debug port** on the same USB-C cable (Windows installs it without a driver). Everything the firmware prints goes there, and the last 4 KB are kept on the board so the boot log can be read later. In the app, click **Board status**: it finds the port, sends `v`, `s` and `l`, and shows the answer; type further commands in the box at the bottom. Any terminal program (PuTTY, Tera Term, Arduino Serial Monitor at any baud rate) works too.

Command	Reply
`s`	uptime, ADC status, streaming flag, USB FIFO fill, PGA gain, both PCM1865 status registers, LRCK edge count, ring-buffer fill, safe-mode flag and crash record
`l`	replay of the boot log (I²C probe, clock configuration, fallbacks taken)
`r`	re-initialise both ADCs and restart capture
`g <dB>`	set the PGA gain of all channels, −12 to +40 dB (default 0); the reply tells you the new full-scale voltage to enter in the app
`v`	firmware version and build date
`h`	help

A healthy board answers `s` and `l` with:

```
AudioAnalyzer_PC fw v2.7 built Oct 28 2026 07:19:57
uptime 483s  adc_ok=1  streaming=1  usb_fifo=7038 bytes (flow control keeps
~half of 14112)  pga=+0.0 dB
safe_mode=0  crash record: resets=0 crumb=0x220 pc=0x00000000 lr=0x00000000
(pc bit31 = stall)
ADC1 @0x4A: state=RUN(0xF) fs=0x03 bckratio=0x44 clkstat=0x00
pll=0x00(unlocked) supply=0x07 clkctrl=0x91
ADC2 @0x4B: state=RUN(0xF) fs=0x03 bckratio=0x47 clkstat=0x11
pll=0x13(locked) supply=0x07 clkctrl=0x0F
LRCK edges in 20 ms: 1920 (expect ~1920)  frames in ring: 6 / 6
---- boot log ----
I2C probe: ADC1(0x4A)=ok ADC2(0x4B)=ok
ADC1 master TDM, LRCK 50%, BCK 256fs: RUN
ADC status: OK
USB started
USB mounted
host set streaming alt 1        <- Windows / the app started the stream
---- end ----
```

Readings that point at a fault: `MISSING` in the I²C probe (address pin or power on that PCM1865), `state=not-run` or a non-zero `clkstat` (clock halt/error bits), `pll=..(unlocked)` on ADC2 (BCK not reaching U4), `LRCK edges: 0` (U3 crystal not oscillating or U3 not in master mode). The firmware tries a fallback clock configuration automatically and reports it.

ADC1 shows `pll=..(unlocked)` and ADC2 `clkstat=0x11` even on a healthy board: ADC1 runs from its crystal without the PLL, and ADC2 reports that it has no SCK because it builds its clocks from BCK. `safe_mode=1` or `resets` above 0 means the firmware crashed and the watchdog restarted it; keep that text, it shows where the firmware stopped. Since v2.6 Board status no longer interrupts the audio, so it is safe to use during a recording.

UART test points (fallback)

If USB itself is the problem (no device at all), the same log is available on the TX/RX/GND test points at 115200 baud 8N1 with a 3.3 V USB-to-serial adapter: adapter RX to TP4 (TX), adapter GND to TP3. The pads are 1.5 mm test points; a probe clip, a pogo pin or a wire held in place is enough, no soldering needed. For the other debugger pads (SWCLK/SWD) a Raspberry Pi Debug Probe can be clipped on the same way, but that is only required for firmware development.

Since firmware v2.4 the UART goes quiet while audio is streaming (printing there caused dropouts); the USB debug port keeps working.

Firmware update

Any time: hold BOOT, tap RESET, release BOOT, copy the new .uf2 onto RPI-RP2. Settings in the app are unaffected. The first line of Board status shows the version; the current one is v2.7. Unplug and replug the board after updating.

Specifications

Parameter	Value
Channels	6 differential speaker-level inputs, each with a hard-wired pass-through output
Full-scale input	≈ 66 Vrms differential at PGA 0 dB (clamp-free to ≈ 45 Vrms); software PGA range −12 to +40 dB available in firmware

Parameter	Value
Input impedance	≈ 52 kΩ per leg (104 kΩ differential)
Frequency response	10 Hz – 20 kHz (AC coupling 4.7 µF; ADC HPF enabled)
ADC	2 × TI PCM1865, 24-bit, 48 kHz, 110 dB SNR class
Resolution floor	about 0.4–0.6 mVrms (≈ −102 dBFS noise floor at 66 Vrms full scale)
Level accuracy	a few % uncalibrated, limited by the meter after calibration
USB	USB 2.0 full speed, USB Audio Class 2.0 (asynchronous), 6 ch × 24-bit × 48 kHz, no driver on Windows 10/11, macOS, Linux; plus a USB serial debug port. The board reports a master volume control that it ignores, so the Windows input-volume slider never scales the data
Power	From USB, < 100 mA at 5 V
Connectors	12 × KF301 5 mm screw terminals (wire up to 1.5 mm²), USB-C
Board	88 × 56 mm, 2-layer FR-4 1.6 mm, 4 × M3 mounting holes
Isolation	None (USB ground = board ground)
Firmware	v2.7 on the RP2040 (pico-sdk 2.1.1 + TinyUSB; rebuild with SDK 2.1.1 only, the USB fix depends on its TinyUSB version), open source in `firmware/`
Software	HU Audio Analyzer for Windows 10/11 (Python 3.10+, PySide6, or a standalone exe from build_exe.bat), WASAPI shared raw mode, CSV logging, 6-ch WAV recording

Reference

Signal path per channel

IN+ and IN− each go through 47 kΩ in series and 4.7 kΩ to analog ground (with the ADC's 10 kΩ input in parallel this divides by 15.7), then a 470 pF capacitor removes RF, a 4.7 µF capacitor blocks DC, and a BAV99 dual diode clamps the ADC pin between the analog 3.3 V rail and ground. The PCM1865 digitizes the pair differentially, so noise picked up equally on both wires cancels.

USB channel map

USB channel	Board input	ADC	ADC input
1	IN1	U3	VIN1P/VIN1M (ADC1L)
2	IN2	U3	VIN2P/VIN2M (ADC1R)
3	IN3	U3	VIN3P/VIN3M (ADC2R)
4	IN4	U3	VIN4P/VIN4M (ADC2L)
5	IN5	U4	VIN1P/VIN1M (ADC1L)
6	IN6	U4	VIN2P/VIN2M (ADC1R)

RP2040 pin map

GPIO	Signal	GPIO	Signal
4	UART1 TX (TP4)	20	I2S BCK (from U3)
5	UART1 RX (TP5)	21	SCK to U4 (unused, driven low)
8	STAT LED	22	I2S LRCK (from U3)
18	I2C1 SDA	23	SCK to U3 (unused, driven low)
19	I2C1 SCL	24	DOUT from U3 (CH1–4)
QSPI	W25Q32 flash + BOOT button on CS	25	DOUT from U4 (CH5–6)

I²C addresses: U3 = 0x4A (MS/AD pin to GND), U4 = 0x4B (MS/AD pin to 3.3 V). Serial format: TDM, 4 slots × 32 bit, BCK = 12.288 MHz, LRCK = 48 kHz with 50 % duty (U3 stays in stand-by as master with the 1-BCK TDM pulse), data offset 1 BCK.

Source files (GitHub: siddukonduru/DigitalAudioAnalyzer)

Path	Content
`firmware\AudioAnalyzer_PC_fw.uf2`	Ready firmware image (v2.7): copy it onto the RPI-RP2 drive
`firmware\src\`, `firmware\CMakeLists.txt`	Firmware source (RP2040, pico-sdk 2.1.1 + TinyUSB)
`pc_app\`	Analyzer app: `app.py`, `dsp.py`, `run.bat`, `build_exe.bat`, `HU_Audio_Analyzer.spec`, `requirements.txt`, `fix_windows_audio.bat`
`.github\README.md`	Repository front page

The board's hardware design files are not part of the repository.